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Improvement of Targeting Efficiency in Chaos Control Using Clustering

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Abstract. In this paper an improved version of the previously presented ECR (Extended Control Regions) targeting method is proposed, where the system data is first pre-processed and subdivided into clusters, and then one artificial neural network is assigned to each such cluster. Furthermore, an analytical criterion for determining the region of the current system state during targeting is introduced, whereas in the original ECR method the region information was hidden in the neural networks. Simulation results on several chaotic systems show that this modified version of the ECR method reduces the average reaching time and in general also the training time of the neural networks.

Keywords: chaos control, OGY method, targeting, clustering, neural networks

1 INTRODUCTION

One of the most popular methods for controlling chaotic dynamics is the OGY method, which is based on the stabilisation of an originally unstable equilibrium point or periodic orbit of the system using local control, as proposed by Ott, Grebogi and Yorke in [1]. The control is achieved by means of small perturbations in one or several control parameters. The fact that no prior knowledge of the system dynamics is needed constitutes one of the major advantages of the OGY approach. On the other hand, the typical drawback of this method is the long waiting time until the system starting from random initial conditions enters the local control region, a small neighbourhood of the chosen target, where the local control can be applied. The remediation of this problem is known under the name of targeting, i.e. the task of steering the chaotic system towards the target. In the literature several targeting methods have been proposed [2-9]. In this work, we present an improved version of Extended Control Regions (ECR) method, our previous approach to targeting [9]. ECR method is based on the idea of identifying those regions of the phase space, starting from which the system can be steered to a close neighbourhood of the target within a few steps using small perturbations in the control parameter. These regions are referred to as the "extended control regions" and are obtained by observing data gathered from the system. Depending on the version of ECR either one or several neural networks are trained to provide the necessary targeting action using data gathered from the system.

In this work a new version of the ECR method (ECR-III) is introduced, where the data corresponding to any extended control region is pre-processed and subdivided into some clusters. Rather than assigning a neural network to each extended control region as in ECR-II, a neural network is assigned to each cluster of each extended control region. Furthermore, unlike the previous versions of ECR, where the region of a current system state was hidden in the trained neural networks and inaccessible to the operator, in ECR-III an analytic criterion is introduced to determine the cluster and the region of the current state.

Section 2 gives the basic information about the OGY method. In Section 3 the ECR approach is explained and in Section 4 the procedures of the different ECR versions are provided. In Section 5 the simulation results are presented for various chaotic systems using different versions of the ECR method. Conclusions are drawn in Section 6.

2 THE OGY METHOD

The OGY method and its modified versions [10-17] do not require any prior knowledge about the system dynamics, rather than that the local linear model of the dynamics in a close neighbourhood of the chosen target is extracted from experimental data. The control is achieved using small perturbations in one or several system parameters.

The basic procedure for constructing the OGY controller can be summarised as follows:

Experimental system data gathered with the nominal parameter values is analysed to detect the unstable equilibrium points and unstable periodic orbits embedded into the strange attractor of the chaotic system, and one of them is chosen as the target.

A second set of experimental data is gathered varying the control parameter(s) randomly about the nominal value(s) within the allowed range Π .

$$\Pi = \{p^i : p_{\text{nom}}^i - \delta p_{\text{max}}^i < p^i < p_{\text{nom}}^i + \delta p_{\text{max}}^i, i=1, \dots, r\} \quad (1)$$

where δp_{max}^i is the maximum allowable parameter change for the i^{th} control parameter and r is the number of control parameters.

Using the second set of experimental data the local linear model in a close neighbourhood of the target is obtained as given in Equation (2).

$$z_{n+1} - z^* = \underline{\underline{A}} (z_n - z^*) + \underline{\underline{B}} (p - p_{\text{nom}}) \quad (2)$$

where z_n denotes the state vector of the system at the n^{th} step, z^* denotes the target state and p_{nom} the nominal parameter vector. $\underline{\underline{A}}$ and $\underline{\underline{B}}$ are the matrices that determine the local linear model.

As soon as the system enters the close neighbourhood of the target the linear model in Equation (2) is used to calculate the necessary variation in the control parameter(s) such that the system is brought to the target at the next control instance.

The major drawback of the OGY method is due to the long waiting time until the system trajectory enters a close neighbourhood of the target.

3 TARGETING VIA EXTENDED CONTROL REGIONS (ECR) METHOD

Depending on the initial conditions and the size of the strange attractor, the reaching time until the system enters the close neighbourhood of the target (the OGY region) may be quite long. In order to reduce this waiting time it is necessary to steer the system towards the OGY region starting from an arbitrary initial point. Since global information about the strange attractor is needed for targeting, local approximations are not useful any more. The proposed Extended Control Regions (ECR) method and its improved versions do not use a complete model of the strange attractor, yet require some information scattered about the whole attractor. Consequently, it is necessary to make use of a tool that has good approximation capabilities over a larger region. All versions of the ECR method maintain the basic assumptions of the OGY control, i.e. no a priori knowledge about system dynamics and control via small perturbations in one or several system parameter(s).

When explaining the procedures for ECR-I, ECR-II and ECR-III, it will be assumed that the necessary processes (Poincaré section for continuous-time systems [11]) have been accomplished to represent the system in the form given in Equation (3) and the target z^* has been chosen.

3.1 ECR-I

The ECR-I method will be presented for an N -dimensional discrete-time chaotic system with r control parameters. Such a system can be represented by Equation (3),

$$z_{n+1} = G(z_n, p), \quad (3)$$

where z_i is the $N \times 1$ state vector at the i^{th} iteration, p_i is the $r \times 1$ control parameter vector applied at the i^{th} iteration and G is an $N \times 1$ dimensional vector function. Assume that this system has an equilibrium point at z^* described by,

$$z^* = G(z^*, p_{\text{nom}}), \quad (4)$$

where p_{nom} is the $r \times 1$ vector of nominal parameter values. Each control variable (parameter) can be changed within its allowable range, i.e. $p \in \Pi$.

The basic idea behind the ECR method is to identify different regions (S_i ; $i = 0, 1, \dots, K$) of the phase space, where the controller will be activated. Hence, the control region of the controller is extended from S_0 (as used in the OGY control) to the union of all S_i 's ($i=0, 1, \dots, K$). These regions can be defined as follows:

Definition 1:

If a system state $\mathbf{z}_n$ obeys (5) then $\mathbf{z}_n \in S_0$.

$$\|\mathbf{z}_n - \mathbf{z}^*\| < \delta, \quad \text{and} \quad \|\mathbf{G}(\mathbf{z}_n, \mathbf{p}) - \mathbf{z}^*\| < \delta \quad \text{with } \mathbf{p} \in \Pi \quad (5)$$

In other words, starting from any $\mathbf{z}_n \in S_0$ the system can be kept within the δ neighbourhood of $\mathbf{z}^*$ at the next step using a $\mathbf{p} \in \Pi$.

Definition 2:

$\mathbf{z}_n \in S_i$ ($i = 1, \dots, K$) if it satisfies (6).

$$\mathbf{z}_n \notin S_0, \quad \text{and} \quad \mathbf{G}^{(i)}(\mathbf{z}_n, \mathbf{p}_n, \mathbf{p}_{n+1}, \dots, \mathbf{p}_{n+i}) \in S_0, \quad \text{with } \mathbf{p}_n, \dots, \mathbf{p}_{n+i} \in \Pi \quad (6)$$

where

$$\mathbf{G}^{(i)}(\mathbf{z}_n, \mathbf{p}_n, \mathbf{p}_{n+1}, \dots, \mathbf{p}_{n+i}) = \mathbf{G}(\mathbf{G}(\dots(\mathbf{G}(\mathbf{z}_n, \mathbf{p}_n)\mathbf{p}_{n+1}), \dots), \mathbf{p}_{n+i}) \quad (7)$$

In other words, S_i is the set of states starting from which the system can be steered to the OGY region (S_0) within i steps. After the extended control regions are determined from the experimental data and an appropriate model describing the system dynamics in all control regions have been obtained this model can be used to steer the system entering S_i to the OGY region (S_0) within i steps.

3.2 ECR-II

ECR-II is an improved version of ECR-I, where the definition of the extended control regions is slightly changed. In order to avoid confusion let us denote the extended control regions employed in ECR-II by T_i . The definitions of the first two regions remain the same, i.e. $T_0 = S_0$ and $T_1 = S_1$.

Definition 3:

$\mathbf{z}_n \in T_i$ if it satisfies (8).

$$\mathbf{z}_n \notin T_{i-1}, \quad \text{and} \quad \mathbf{G}(\mathbf{z}_n, \mathbf{p}_n) \in T_{i-1}, \quad \mathbf{p}_n \in \Pi, \quad i=2, 3, \dots, K \quad (8)$$

In other words, T_i is the set of states starting from which the system can be steered to T_{i-1} in one step. This modification allows separate modelling of the different control regions. When the system enters T_i it can be steered to T_{i-1} in the next step using the model of T_i .

3.3 ECR-III

Applications of the ECR approach to various chaotic systems have shown that control regions do not have to be connected or simply connected. Furthermore, because data corresponding to higher control regions tend to be scattered and because finite experimental data are used, sufficient information about the control regions may not be available.

The main idea in the next improved version ECR-III is to try to extract more reliable models of system dynamics from the available data by considering data clusters separately instead of trying to interpolate between scattered region data.

Consequently, in ECR-III the control regions (T_i 's) in ECR-II method are clustered into smaller sub-regions (clusters) using a simple clustering algorithm, which will be explained in following sections. The system dynamics within each cluster of each control region is modelled separately. The loci of the clusters are represented in terms of the cluster means and the normalised cluster variances. Such a representation renders an analytical criterion possible for determining in which cluster a current system state lies.

4 APPLICATION OF ECR USING NEURAL NETWORKS

The modelling in all different versions of the ECR method has been realised using neural networks. Because of its high modelling and approximation capability, radial basis functions based neural networks have been preferred. Radial basis functions can be viewed as a smooth transition between fuzzy logic and neural networks, which are very useful tools in soft computing.

4.1 ECR-I Method

Application of the procedure associated with the off-line training phase of the ECR-I method can be described as follows:

The system is run while changing the control parameter(s) within the allowable range randomly, until some predetermined termination criterion is satisfied. The data set obtained from such a run contains the successive state pairs (z_n, z_{n+1}) and the corresponding parameter values (p_n) 's.

The data are analysed and the control regions $(S_i, i = 0, \dots, K)$ are determined using Definitions 1 and 2.

A single neural network with two input ports and one output port is trained using the data set as shown in Fig. 1. If not all state variables are measurable and delay coordinates [11,18] have to be used, a third input port needs to be used, where the previous parameter values $(p_{n-1}, p_{n-2}, \text{etc.})$ can be fed to the neural network.

The neural network trained as described above can be used for the control action according to the following procedure:

In the control phase, the current state z_n is fed to port 1 and the target z^* to port 2. The neural network provides $\tilde{p}_n$, a candidate control parameter vector to be applied at step n .

The candidate parameter vector $\tilde{p}_n$ may be meaningless, if the current state z is in none of the control regions $(S_i, i = 0, \dots, K)$ or if the neural network is insufficient in modelling the control relationship for the current state. These cases correspond to a situation, where the neural network is being used either outside their training region or beyond their approximation capacity. Assuming that such a situation can be recognised by a parameter vector that lies outside the allowable parameter range, the following correction is made: If $\tilde{p}_n \in \Pi$, then $p_n = \tilde{p}_n$ else if $\tilde{p}_n \notin \Pi$, then $p_n = p_{nom}$.

The control parameter vector obtained in this manner is used in step n . When this procedure is followed, the system is supposed to go to: z^* if $z_n \in S_0$, or S_{i-1} if $z_n \in S_i$, or some undefined state if $z_n \notin S_i, \forall i = 0, \dots, K$.

In ECR-I, there is no direct knowledge about the location of the current state. The neural network trained by sufficiently large training data simply provides the necessary change in the control parameter.

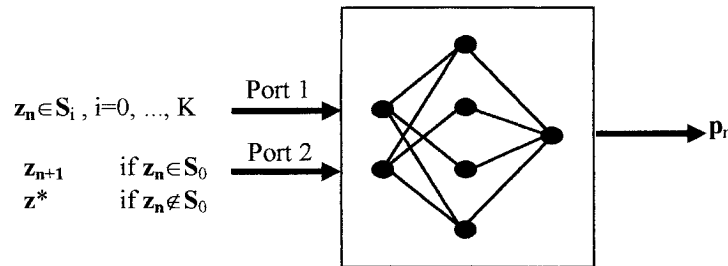


FIGURE 1. Training phase of NN using ECR-I method

4.2 ECR-II Method

The improvement in ECR-II is the usage of a separate neural network for each control region (T_i) 's. Such a division of tasks naturally results in better modelling performance. In this case, the training data belonging to each control region (T_i) 's have to be classified sequentially according to Definition 3, as opposed to ECR-I. The off-line training phase for ECR-II can be summarised as follows:

The system is run while changing the control parameter(s) within the allowable range randomly, until some predetermined termination criterion is satisfied. The data set obtained from such a run contains the successive state pairs (z_n, z_{n+1}) and the corresponding control parameter values p_n 's.

The training data belonging to T_0 are extracted according to Definition 1 ($T_0 = S_0$). The neural network for T_0 (NN_0) is trained using these data. NN_0 trained in this manner is supposed to provide the necessary p values for local control.

The training data belonging to T_1 are extracted according to Definition 2 ($T_1 = S_1$). The neural network for T_1 (NN_1) is trained using these data. NN_1 trained in this manner is supposed to produce the necessary p_n values that will take the system from $z_n \in T_1$ to somewhere within T_0 .

The training data belonging to T_i ($i = 2, 3, \dots, K$) are extracted using NN_{i-1} as described in the flowchart in Fig. 2. Any NN_i is trained using the training data belonging to T_i . NN_i is supposed to provide the necessary p values that will take the system from $z_n \in T_i$ to somewhere within T_{i-1} .

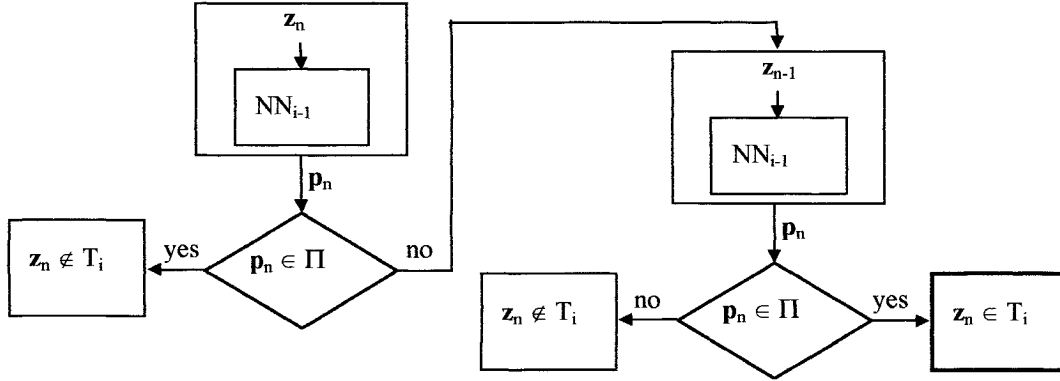


FIGURE 2. Flowchart describing the data extraction phase of NN_i using ECR-II method

Once all neural networks (NN_i , $i = 0, \dots, K$) have been trained, they can be used for targeting and local control as follows: The current state is fed sequentially to each neural network (NN_i , $i = 0, \dots, K$) starting from NN_0 , until one of them produces a $p_n \in \Pi$.

In ECR-II, the location of the current state of the system is determined by the neural networks in the following manner: The current state is fed to the neural networks belonging to different regions starting from the one with the highest index i . This iterative process stops when one of the neural networks produces a control parameter change within the allowable range. The region with the largest index, for which the neural network produces an acceptable parameter change, is regarded as the region of the current state. If none of the neural networks produces an acceptable parameter change, the current state is assumed to be in none of the control regions and no targeting action is taken. Actually, the criterion described above may eventually produce wrong results concerning the region of the current state but it still provides a correct result for an overwhelming majority of cases, hence a reasonable improvement in the average waiting time.

4.3 ECR-III Method

In this new version the control regions are obtained in the same manner as in ECR-II. After the data belonging to any control region has been obtained they are made subject to a simple clustering algorithm. The j^{th} data cluster corresponding to the i^{th} control region is denoted by C_{ij} and is assigned a neural network NN_{ij} . Each NN_{ij} is trained using the data corresponding to C_{ij} to provide the necessary control parameter values that will take the system to a cluster of the $(i-1)^{\text{st}}$ control region, $C_{(i-1)k}$ at the next step.

4.3.1 The Clustering Algorithm

The clustering algorithm used for determining any C_{ij} can be explained as follows:

An arbitrary data point in a control region, T_i , is taken as a centre of a hyper-sphere with a radius r and all data points that lie in that hyper-sphere are labelled with same index, which is equal to 1 ($j = 1$) at the beginning. Next, another data point in same control region is taken as a new centre of the hyper-sphere. At this step, there are two possible situations:

If this new data point has a label, the clustering continues with another point. If not, a hyper-sphere is considered, the centre of which is the new data point. In this hyper-sphere a labelled datum is searched for. If a labelled datum can be found in that hyper-sphere, all new data points in that hyper-sphere are labelled with the same index as the pre-labelled data point. If not, this new data point is labelled by an incremented index number, $j = j + 1$.

The clustering algorithm is terminated when all data points in the control region under consideration are exhausted.

In this algorithm, the radius of the hyper-sphere, r is obtained from the region data by visual analysis of the histogram of inter-data distances as the first minimum of the histogram.

After the clusters, C_{ij} are obtained, the “Normalized Covariance Matrix” of the cluster C_{ij} denoted by N_{ij} and the mean of cluster C_{ij} denoted by μ_{ij} are calculated.

This modification introduced in ECR-III allows the determination of the current state location in an analytical manner as follows:

For the current state z_n , “Normalized Mahalanobis Distance” (NMD) from all C_{ij} ’s are calculated using Equation (9),

$$(\text{NMD})_{ij}^2(z_n) = (z_n - \mu_{ij})^T \underline{N}_{ij} (z_n - \mu_{ij}) \quad (9)$$

where $\underline{N}_{ij}$ is the “Normalized Covariance Matrix” of cluster C_{ij} , and the μ_{ij} is the mean of C_{ij} .

The cluster C_{ij} , which has the minimum value of Normalised Mahalanobis Distance NMD_{ij} from the current state, is chosen and the corresponding neural network NN_{ij} is activated to obtain the necessary control parameter(s) for targeting. If NN_{ij} produces a control parameter change within the allowable range, that cluster is regarded as the region of the current state. If the neural network produces an unacceptable parameter change the current state is assumed to be in none of the control regions and no targeting action is taken.

5 SIMULATION RESULTS AND COMPARISON OF DIFFERENT ECR VERSIONS

In this study, all three versions of ECR have been simulated on various chaotic systems both under noiseless and noisy conditions. In order to determine the robustness of the methods with respect to measurement noise, some random noise proportional to the root mean square value of the measured data (state measurement) is added to the state measurements.

The percentage of the maximum tolerable (i.e. allowing the stabilisation of the target) measurement error is determined separately for each chaotic system and each control method (OGY, ECR-I, ECR-II and ECR-III). Furthermore, the average reaching time is determined for each system and each method under noisy conditions, where the measurement noise is taken as the minimum among the maximum tolerable percentage measurement

TABLE 1. Investigated Chaotic Systems and Parameters

SYSTEMS and EQUATIONS	Equilibrium Point (z^*)	Nominal Parameter Value(s)	Max. Allowable Parameter Change	RMS value of the Measured State	Radius of the OGY Region(δ)
Logistic Map $x_{n+1} = px_n(1-x_n)$	$x^* = 0.7435$	$p_{\text{nom}} = 3.9$	$\delta p_{\text{max}} = 0.10$	$x_{\text{rms}} = 0.6608$	$\delta = 0.01$
Henon Map $x_{n+1} = p + 0.3y_n - x_n^2$ $y_{n+1} = x_n$	$x^* = 0.8717$ $y^* = 0.8717$	$p_{\text{nom}} = 1.37$	$\delta p_{\text{max}} = 0.03$	$x_{\text{rms}} = 1.327$	$\delta = 0.02$
Lorenz System $\dot{x} = \sigma(y - x)$ $\dot{y} = (\rho x - y - xz)$ $\dot{z} = (xy - \beta z)$	<u>Real Coord.</u> $y_{\text{PSS}} = 8.4853$ $x^* = 14.2387$ $z^* = 39.7934$	$\sigma_{\text{nom}} = 10$ $\rho_{\text{nom}} = 28$ $\beta_{\text{nom}} = 8/3$	$\delta \sigma_{\text{max}} = 0.30$ $\delta \rho_{\text{max}} = 0.84$ $\delta \beta_{\text{max}} = 0.08$	$x_{\text{rms}} = 13.43$	$\delta = 0.30$
	<u>Delay Coord.</u> $y_{\text{PSS}} = 8.4853$ delay time (T) T = 100 ms dc1* = -3.0988 dc2* = -3.2911 dc3* = -3.8340	$\sigma_{\text{nom}} = 10$ $\rho_{\text{nom}} = 28$ $\beta_{\text{nom}} = 8/3$	$\delta \sigma_{\text{max}} = 0.30$ $\delta \rho_{\text{max}} = 0.84$ $\delta \beta_{\text{max}} = 0.08$	$y_{\text{rms}} = 2.675$	$\delta = 0.30$
Duffing Oscillator $\dot{x} = y$ $\dot{y} = -x - x^3 - dy + f \cos(wz)$ $\dot{z} = 1$	<u>Delay Coord.</u> $x_{\text{PSS}} = 1$ delay time (T) T = 2376 ms dc1* = 2.3935 dc2* = 0.5839	$d_{\text{fixed}} = 0.2$ $f_{\text{nom}} = 36$ $w_{\text{fixed}} = 0.661$	$\delta f_{\text{max}} = 0.36$	$x_{\text{rms}} = 2.018$	$\delta = 0.05$

TABLE 2. Comparison of the OGY Control and Various ECR Methods for Different Chaotic Systems

LOGISTIC Map	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III
# of activation regions	0	2	1		2		3		4		5		6	
Training time (flops)	$0.16 \cdot 10^6$	$35 \cdot 10^6$	$2.89 \cdot 10^6$	$3.56 \cdot 10^6$	$4.05 \cdot 10^6$	$3.73 \cdot 10^6$	$5.33 \cdot 10^6$	$3.96 \cdot 10^6$	$6.72 \cdot 10^6$	$4.07 \cdot 10^6$	$6.76 \cdot 10^6$	$4.18 \cdot 10^6$	$8.06 \cdot 10^6$	$4.21 \cdot 10^6$
Av. reach. t. (without noise)	143.7	50.87	51.31	58.571	46.37	53.027	13.75	8.575	9.32	5.753	9.23	5.316	9.14	5.523
Av. reach. t. (with 0.30% noise)	156.8	52.64	52.96	58.153	46.99	52.797	14.16	8.694	10.96	5.894	11.26	5.418	11.25	5.747
HENON Map	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III
# of activation regions	0	3	3		4		5		6		7		8	
Training time (flops)	$0.92 \cdot 10^6$	$601 \cdot 10^6$	$33 \cdot 10^6$	$28.4 \cdot 10^6$	$35 \cdot 10^6$	$29.2 \cdot 10^6$	$42 \cdot 10^6$	$29.8 \cdot 10^6$	$48 \cdot 10^6$	$30.5 \cdot 10^6$	$56 \cdot 10^6$	$31.0 \cdot 10^6$	$57 \cdot 10^6$	$31.6 \cdot 10^6$
Av. reach. t. (without noise)	767.9	108.73	107.18	121.76	101.72	89.56	59.89	57.95	30.06	27.89	21.94	17.17	22.50	15.39
Av. reach. t. (with 0.22% noise)	837.8	134.9	126.00	108.58	104.82	82.56	60.25	57.59	39.29	26.52	31.00	15.51	30.65	13.64
LORENZ Syst. (real coord.)	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III
# of activation regions	0	1	1		2		3		4		5		6	
Training time (flops)	$0.21 \cdot 10^6$	$96 \cdot 10^6$	$443 \cdot 10^6$	49.910^6	$454 \cdot 10^6$	$50.6 \cdot 10^6$	$455 \cdot 10^6$	$51.1 \cdot 10^6$	$455 \cdot 10^6$	$51.4 \cdot 10^6$	$456 \cdot 10^6$	$51.5 \cdot 10^6$	$456 \cdot 10^6$	$51.5 \cdot 10^6$
Av. reach. t. (without noise)	15.09	6.64	6.14	6.80	6.43	6.44	6.45	6.43	6.57	6.39	6.45	6.42	6.66	6.42
Av. reach. t. (with 0.08% noise)	18.91	6.81	6.43	7.93	6.02	7.79	6.34	7.49	5.93	6.82	5.92	6.65	6.86	6.59
LORENZ Syst. (delay coord.)	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III
# of activation regions	0	1	1		2		3		4		5		6	
Training time (flops)	$0.24 \cdot 10^6$	$5.91 \cdot 10^{10}$	$4.81 \cdot 10^9$	$0.84 \cdot 10^9$	$4.84 \cdot 10^9$	$0.87 \cdot 10^9$	$4.86 \cdot 10^9$	$0.88 \cdot 10^9$	$4.87 \cdot 10^9$	$0.89 \cdot 10^9$	$4.88 \cdot 10^9$	$0.89 \cdot 10^9$	$4.89 \cdot 10^9$	$0.89 \cdot 10^9$
Av. reach. t. (without noise)	35.32	18.31	17.92	16.21	13.62	14.90	12.83	14.02	12.76	12.23	12.73	12.61	12.73	12.47
Av. reach. t. (with 0.36% noise)	34.20	20.08	18.93	15.37	15.31	15.48	14.21	12.94	15.24	11.67	15.26	11.70	15.22	11.25
DUFFING Osl. (delay coord.)	OGY	ECR-I	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III	ECR-II	ECR-III
# of activation regions	0	1	1		2		3		4		5		6	
Training time (flops)	$0.48 \cdot 10^6$	$5.03 \cdot 10^{10}$	$1.08 \cdot 10^9$	$0.25 \cdot 10^9$	$1.12 \cdot 10^9$	$0.26 \cdot 10^9$	$1.14 \cdot 10^9$	$0.26 \cdot 10^9$	$1.15 \cdot 10^9$	$0.26 \cdot 10^9$	$1.15 \cdot 10^9$	$0.26 \cdot 10^9$	$1.16 \cdot 10^9$	$0.27 \cdot 10^9$
Av. reach. t. (without noise)	107.55	42.94	39.81	42.86	16.40	21.64	13.20	11.23	13.98	11.56	14.03	12.91	13.68	15.47
Av. reach. t. (with 0.27% noise)	116.70	48.40	51.40	60.67	27.32	36.10	18.65	17.64	18.97	17.40	18.47	21.61	17.83	23.40

errors corresponding to the three different methods. The average reaching time is calculated as the average of the time it takes the system to enter the OGY region starting from many different initial conditions (500 different initial conditions for discrete-time systems, 100 for continuous-time systems). Delay coordinates mentioned in the table refer to the method employed when in a continuous-time system only a single state can be measured. In this case, a new coordinate system is constructed using delayed values of the measurable state [18]. Results obtained for the chaotic systems in Table 1 are summarised in Table 2.

Table 2 summarizes the performances of OGY, ECR-I, ECR-II and ECR-III methods for the chaotic systems given in Table 1. The OGY method implemented here corresponds to the basic OGY algorithm where the coefficient matrices of the local linear model (Equation (2)) are obtained from the training data by the least square error approach. For all versions of the ECR method, Table 2 presents the results for various numbers of control regions. The maximum tolerable percentage noise is defined as the maximum amount of uniformly distributed noise (as a percentage of the rms value of the measured state), which can be added to the state measurements still allowing the stabilization of the target.

The table also shows the average reaching times under noisy conditions, where the minimum of the maximum tolerable noises corresponding to different strategies have been used.

From Table 2 it can be observed that for the optimal number of extended control regions ECR-III provides better results than ECR-II both with respect to average reaching time and training time both with and without measurement noise. Although in ECR-III more neural networks (one for each cluster of each control region) are used the overall training decreases because the data in each cluster is much more homogeneous than the data in the control regions used in ECR-II.

Furthermore, although in ECR-III the overall hyper-volume of all extended control regions (union of all clusters) has decreased as compared to ECR-II the average reaching time is reduced. This phenomenon can be explained as follows: The experimental data do not provide any reliable information about the state space regions, which remain between the clusters. Nevertheless, ECR-II produces some control activity in those regions thanks to the interpolation property of the neural networks. When such interpolations fail to capture the actual system dynamics the waiting time may become longer than it would be without any control activity. ECR-III, by avoiding control action in such inter-cluster regions, eliminates this problem and results in a shorter average waiting time. Another reason behind the better average reaching time performance of ECR-III is the criterion for determining the region/cluster of the current system state. While in ECR-II the determination of the region of the current state is merely based on whether the neural network of a region produces an acceptable parameter change, ECR-III determines the region/cluster of the current state using a more reliable distance criterion. Again, such a modification reduces the probability of wrong identification of the current state loci, thus avoiding wrong control activity and resulting long waiting time.

6 CONCLUSION

In this work, the three different versions of the neural network based ECR method have been presented. These versions differ from each other in the way extended control regions are defined and modeled, and the way the region of a current system state is determined. The simulation results show that ECR-III gives the best performance both in terms of average reaching time and off-line training time of the neural networks as a result of splitting the region data into sub-clusters. In ECR-III a very simple clustering algorithm has been preferred although more sophisticated clustering algorithms may improve the average reaching time even further on cost of increasing the training time.

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